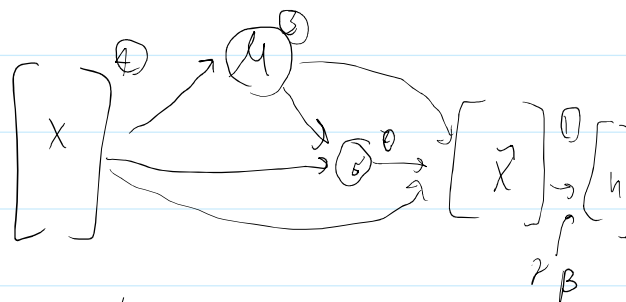


$$\mu = \frac{1}{m} \sum x_i$$

$$\sigma^2 = \frac{1}{m-1} \sum (x_i - \mu)^2$$

$$\hat{x}_i = \frac{x_i - \mu}{\sqrt{\sigma^2 + \epsilon}}$$

$$y_i = \gamma \hat{x}_i + \beta$$



$$\frac{dL}{dy_i} \rightarrow \frac{dL}{dx_i}$$

$$(1) \frac{dL}{d\hat{x}_i} = \frac{dL}{dy_i} \cdot \gamma$$

$$(2) \frac{dL}{d\sigma^2} = \sum_i \frac{dL}{d\hat{x}_i} \frac{d\hat{x}_i}{d\sigma^2} = \gamma \sum_i \frac{d}{d\sigma^2} [(x_i - \mu)(\sigma^2 + \epsilon)^{-1/2}] \frac{dL}{dy_i}$$

$$= -\frac{1}{2} \gamma \sum_i \frac{dL}{dy_i} (x_i - \mu)(\sigma^2 + \epsilon)^{-3/2}$$

$$(3) \frac{dL}{d\mu} = \sum_i \left(\frac{dL}{d\hat{x}_i} \frac{d\hat{x}_i}{d\mu} \right) + \left(\frac{dL}{d\sigma^2} \frac{d\sigma^2}{d\mu} \right)$$

$$\frac{d\hat{x}_i}{d\mu} = \frac{d}{d\mu} \left[\frac{x_i - \mu}{\sqrt{\sigma^2 + \epsilon}} \right] = -(\sigma^2 + \epsilon)^{-1/2} \quad \left| \quad \frac{d\sigma^2}{d\mu} = \frac{d}{d\mu} \left[\frac{1}{m-1} \sum_{i=1}^m (x_i - \mu)^2 \right] \right.$$

$$\frac{dL}{d\mu} = -(\sigma^2 + \epsilon)^{-1/2}$$

$$= \frac{-2}{m-1} \sum (x_i - \mu)$$

$$= \frac{-2}{m-1} (\sum x_i - \sum \mu)$$

$$= \frac{-2}{m-1} (m\mu - m\mu) = 0$$

$$\mu = \frac{1}{m} \sum x_i \quad \leftarrow \begin{array}{l} \text{# of neurons} \end{array}$$

$$(4) \frac{dL}{dx_i} = \frac{dL}{d\hat{x}_i} \cdot \frac{d\hat{x}_i}{dx_i} + \frac{dL}{d\mu} \frac{d\mu}{dx_i} + \frac{dL}{d\sigma^2} \frac{d\sigma^2}{dx_i}$$

ker eq

$$\frac{d\hat{x}_i}{dx_i} = \frac{d}{dx_i} \left[\frac{x_i - \mu}{\sqrt{\sigma^2 + \epsilon}} \right] = (\sigma^2 + \epsilon)^{-1/2} \quad \left| \quad \frac{d\mu}{dx_i} = \frac{d}{dx_i} \left[\frac{1}{m} \sum x_i \right] = \frac{1}{m} \right.$$

$$\frac{\partial \sigma^2}{\partial x_i} = \frac{\partial}{\partial x_i} \left[\frac{1}{n-1} \sum_j (x_j - \mu)^2 \right] = \frac{2}{n-1} (x_i - \mu)$$

Bring eq back

$$\begin{aligned}
 &= \left(\frac{\partial L}{\partial \mu_i} \cdot \gamma (\sigma^2 + \epsilon)^{-1/2} \right) + \left(- \sum_j \frac{\partial L}{\partial \mu_j} \cdot \gamma \cdot (\sigma^2 + \epsilon)^{-1/2} \right) \cdot \frac{1}{n} \\
 &+ \left(- \frac{1}{2} \gamma \sum_j \frac{\partial L}{\partial \mu_j} \cdot (x_j - \mu) (\sigma^2 + \epsilon)^{-3/2} \right) \left(\frac{2}{n-1} (x_i - \mu) \right) \\
 &\downarrow \\
 &= (\sigma^2 + \epsilon)^{-1/2} \left(\frac{\partial L}{\partial \mu_i} \cdot \gamma \right) + (\sigma^2 + \epsilon)^{-1/2} \left(- \frac{1}{n} \sum_j \frac{\partial L}{\partial \mu_j} \gamma \right) + (\sigma^2 + \epsilon)^{-1/2} \left(- \gamma \sum_j \frac{\partial L}{\partial \mu_j} \frac{x_j - \mu}{\sqrt{\sigma^2 + \epsilon}} \right) \left(\frac{1}{n-1} \cdot \frac{x_i - \mu}{\sqrt{\sigma^2 + \epsilon}} \right) \\
 &= \frac{(\sigma^2 + \epsilon)^{-1/2}}{n} \left[n \gamma \left(\frac{\partial L}{\partial \mu_i} \cdot \sum_j \frac{\partial L}{\partial \mu_j} \gamma - \gamma \left(\sum_j \frac{\partial L}{\partial \mu_j} \tilde{x}_j \right) / \left(\sum_{j=1}^n x_j \right) \right) \right] \quad \tilde{x}_j = \downarrow \\
 &= \frac{\gamma (\sigma^2 + \epsilon)^{-1/2}}{n} \left[n \frac{\partial L}{\partial \mu_i} - \sum_j \frac{\partial L}{\partial \mu_j} - \frac{n}{n-1} \tilde{x}_i \sum_j \frac{\partial L}{\partial \mu_j} \tilde{x}_j \right]
 \end{aligned}$$